

CSC360 - Computer Graphics and Digital Image Processing

3/9/26 Reflection

Topic: Console Printing, ASCII Trees, Graphics, SSH, JavaFX, FXML and Testing

Reflection

This lecture has shown me the distinction between printing of the textual structure and graphical structure. I learned that ASCII tree uses special characters such as `*`, `/`, `\`, `|`, `-`, `|` and `L` in order to create the tree, but for the creation of graphics one should use the graphics libraries such as JavaFX or Graphics2D. This helps one to create or print something if he is asked to do it.

Another concept that I have learned is the printing of hierarchical structures, like the directory tree, in the console. The most difficult thing in this process is the alignment of the structure especially in case when there are several levels in the list and the match occurs at several points of two lists. Instead of considering that there is the match on the same level, their indexes can be stored and the position of the arrow can be calculated.

Yet another topic we considered was SSH (Secure Shell). I knew that a computer that does not have GUI could be accessed remotely via command line interface through SSH. This command – `ssh username@IP_address` – will connect to the remote computer and make it possible to execute commands on it.

We learned about JavaFX, FXML and splash screens. FXML enables describing the user interface in a separate document from Java code. A splash screen is a screen which appears during initialization of an application and shows the application name or its logo.

Key Concepts Learned

- ASCII Tree – A tree-like object made up of characters.
- Printing and Drawing – Printing refers to showing characters in a console, whereas drawing refers to making actual drawings.
- Console – A text interface through which inputs and outputs can be made.
- Directory Structure – A hierarchical tree where each object is a parent of others with objects as its children; can be depicted with |, ├──, and └─ symbols.
- Visual Mapping – Connecting corresponding elements in two lists by drawing arrows.
- Position/ Index Comparison – Useful when elements to match are at different indexes in the two lists.
- SSH – A technique for connecting securely and remotely to another computer using the command line interface.
- JavaFX – A platform for building user interfaces using Java.
- FXML – XML language to specify UI in JavaFX.
- Splash Screen – The starting/loading page of an application.
- Unit Testing – Testing of small components of the software like functions and classes.
- Acceptance Testing – Testing of entire application according to specifications.

Key learning

What I learned in this regard was that depending on the type of output that was desired, a particular representation technique could be chosen. While basic text-based output could be obtained using console printing and ASCII art, graphical output required the use of a GUI and graphics libraries. Another important lesson was the fact that systems could be controlled remotely through SSH and the use of JavaFX, FXML, and testing.